

In this lab, a bridge crane and a tower crane were used to analyze the effects of various shaping methods on residual vibration.

Bridge Crane

The first part of the lab was performed on a bridge crane setup. The crane was tested at fixed cable length of 800mm with 3 different pulse times of 1100ms, 1600ms, and 2000ms using 3 different shaping methods. The shaping methods used were unshaped, a ZV shaper, and a robust ZVD shaper. For all three pulse times, both shapers drastically decreased residual vibration by timing the pulses to cancel out the vibration.

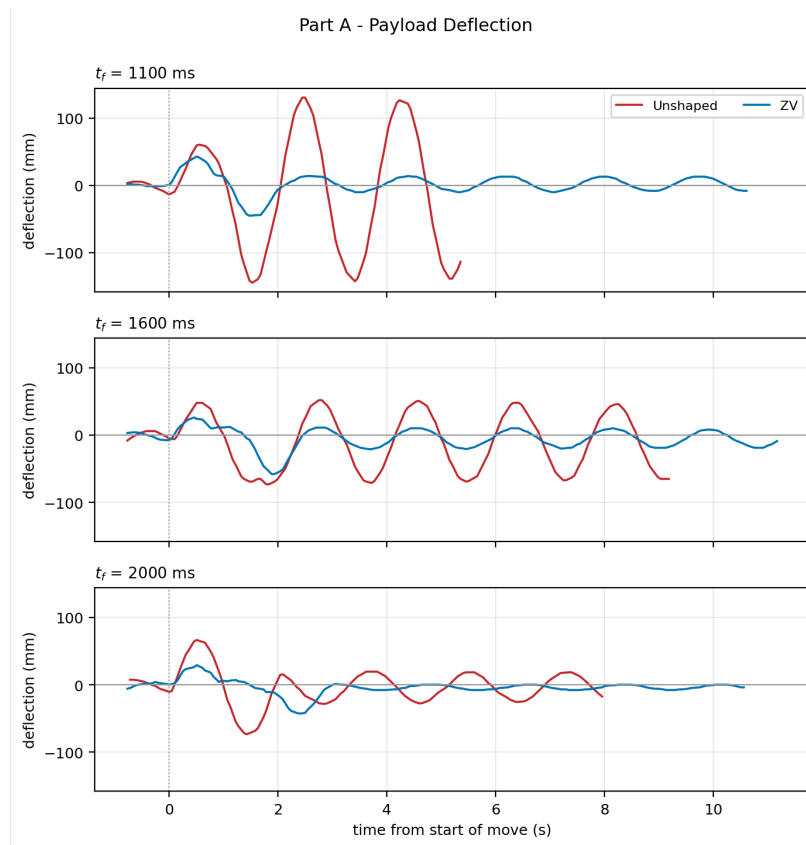


Figure 1 – This figure plots the deflection of the payload in millimeters for all three different pulse times versus the time from the start of move in seconds. The red curves represent the unshaped commands and the blue curves represent the ZV shaped commands.

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Figure 1 shows the comparison between the unshaped and the ZV shaped response. During the impulse the two responses are very similar due to the payload movement, but the second impulse of the shaped response significantly decreases the residual vibration. For all three trials the residual vibration is significantly lower when the ZV shaper is used. Equation (1) shows the relationship between the length of the crane and the time period of oscillation. The pulse time of 1100ms creates the worst residual oscillation because the pulse aligns with the phase of the pendulum adding more energy into the system.

$$T = 2\pi \sqrt{\frac{l}{g}} \quad (1)$$

To test the robustness of the ZV shaper, the bridge crane was run for $t_i = 1500$ ms at cable lengths of 0.6, 0.9, and 1.2 m. At each of these lengths, an unshaped test was taken along with a ZV shaper and a ZVD shaper to test the robustness of the ZV shaper compared to no shaper and compared to the typically more robust ZVD shaper.

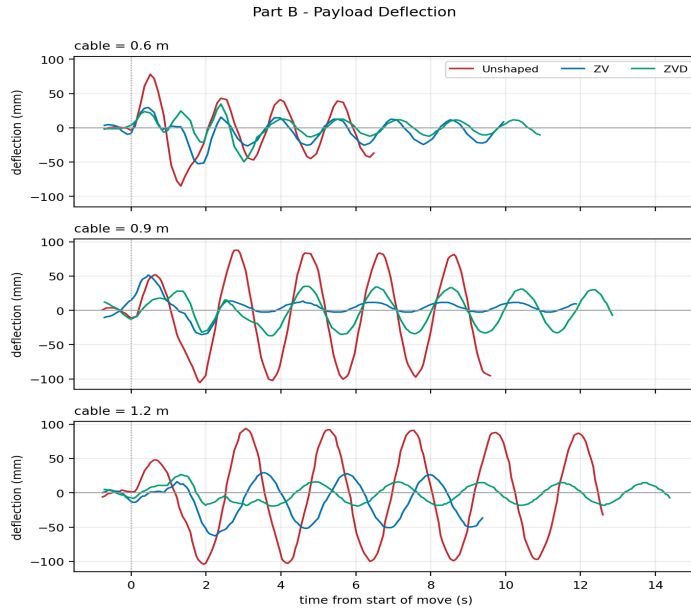


Figure 2 – This figure plots the deflection of the payload in millimeters for all three different pulse times verses the time from the start of move in seconds. The red curves represent the unshaped commands, the blue curves represent the ZV shaped commands, and the green curves represent the robust ZVD shaped commands.

As seen above in Figure 2, while the ZV shaper performs better at the 0.9m cable length it is designed around, the ZVD shaper performs better at the other cable lengths. This shows how the ZV shaper can be better for a well understood system, but the ZVD shapers robustness makes it the better choice where systems parameters may deviate from theoretical calculations.

Tower Crane

In the tower crane experiment¹, four different types of input shapers were compared to analyze which reduced the residual oscillations present after a slewing motion occurred. This was tested with the trolley at different positions: 500, 700, and 900 millimeters. The experiment results demonstrated that the input shaping substantially reduced the residual payload oscillations that were generated by the tower crane slew motion. As predicted, the unshaped trials produced the largest oscillations, and the tangential response increased noticeably as the radius of the trolley increased. This increase in the residual oscillations with the radius is consistent with the rotational motion of the crane, since the tangential velocity and acceleration of the trolley should increase with increasing radial distance from the axis of rotation. The single-mode ZV shaper reduced tangential oscillations but was less effective when it came to reducing radial motion. This limitation became more apparent at larger trolley radii.

The robust and two-mode shapers provided a significantly larger suppression in both the radial and tangential directions. Of the four designs tested, unshaped, ZV, ZVD, and two-mode ZV, the robust shaper produced the lowest sustained oscillations across the three tested trolley positions. The two-mode shaper was also effective and showcased the benefit of accounting for the crane's double pendulum, multimodal dynamics. The robust shaper was deemed best overall due to the mean residual RMS value being the lowest of all the tested shapers, and this can be seen in **FIGURE** __ below. However, the two-mode shaper reduced the oscillations of the higher frequency modes more efficiently, although it allowed a larger transient motion before settling (see **Figure** __ below). This suggests that both frequency, robustness, and multi-mode suppression are important aspects of designing an input shaper for the tower crane which may have different optimal shapers depending on priority.

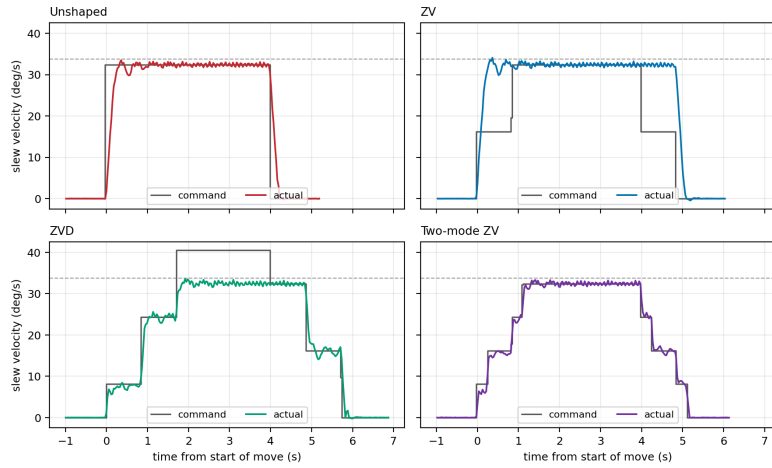
Within the context of this experiment, the robust ZVD shaper is the best, since it has lower residual error than all other shapers for both tangential and radial swing at all lengths.

¹ For the tower crane experiment, the lab manual had a typo at the time of data collection. Two lengths were listed for the crane cable length: 1.0 and 0.6 meters. Our group went with 0.6 meters and were later informed that the correct option was one meter. After discussing with the TA, we were told we may use the data we had already collected, so long as a note was made in the report.

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While the two stage ZV shaper suppressed the higher order frequency oscillations, the swing caused by the lower frequency oscillations is still significant and our goal is to dampen all of the oscillations without preference towards specific frequencies. Not only is the ZVD shaper the most robust, it also has the lowest swing error for every trial, making it the clear best choice.

Tower - Commanded vs Actual Slew Velocity (trolley 500 mm)



Tower Crane Residual Swing

